HOUSE PRICE ESTIMATION

A supervised machine-learning workflow for estimating property prices

LINEAR REGRESSION

500 SYNTHETIC RECORDS

PROJECT BRIEF

PROGRAM

Machine Learning | IIOT-4

COURSE

BCA (Artificial Intelligence), Section A

MENTOR

Harsh Shrivastav

OUTPUT

Estimated price in INR

Educational demonstration - not a formal property valuation

PROJECT TEAM

The team behind the project

Student names and IDs retained from the original presentation.

TEAM LEADER	**SHAGUN BAJPAI**	Student ID: BCAI2025220 | Roll No.: 2510208005

01	ANKIT PRAJAPATI	BCAI2025184 | 2510206024
02	TANISHKA AGARWAL	BCAI2025195 | 2510206172
03	SACHIN KUMAR	BCAI2025072 | 2510206141
04	DEEPANSHI CHAUDARY	BCAI2025121 | 2510206051
05	SAMRIDDHI SHUKLA	BCAI2025170 | 2510206146
06	VEDIKA SHARMA	BCAI2025181 | 2510206186
07	TANYA RATHOR	BCAI2025193 | 2510206173
08	VEDIKA PATHAK	BCAI2025200 | 2510206185
09	ANMOL SINGH	BCAI2025204 | 2510206027
10	YOGESH KUMAR	BCAI2025206 | 2510206189
11	RITIK KUMAR	BCAI2025221 | 2510289085
12	HARSIMAR SINGH	BCAI2026217 | 2510208003
13	VANSH DIWAKAR	BCCC2025005 | 2510208006
14	ABHISHEK KUMAR	BCCC2025008 | 2510208001
15	KRISHNA PRAJAPATI	BCCC2025009 | 2510208004
16	FAIZAN KHAN	BCC2025010 | 2510208007

A data-driven estimate from property features

House prices vary with characteristics and market context; a model can learn patterns from labelled examples.

01

Problem

Manual estimates can differ. Similar-sized homes may have different prices due to location, age, layout and market conditions.

02

Approach

Supervised regression learns from records containing property inputs and known listed prices, then predicts unseen cases.

PROJECT OBJECTIVES

Prepare

Inspect and clean a structured dataset.

Explore

Review feature and price patterns.

Model

Encode categories and train Linear Regression.

Evaluate

Measure held-out prediction errors and discuss limits.

EXPECTED OUTCOME

A reproducible demonstration that returns an estimated price with evaluation metrics.

A reproducible dataset for a controlled experiment

The report describes 500 synthetic records. Their purpose is to demonstrate the workflow, not represent a live market.

500	records
400	training rows
100	test rows
42	random state

INPUT FEATURES

Area_sqft | Bedrooms | Bathrooms | Age_years | Location_Category

TARGET

Price_INR

DATA UNDERSTANDING & EDA

Load CSV -> inspect rows, types and summary statistics -> check missing values, duplicates and valid numeric fields -> use descriptive statistics and area-versus-price plots to identify trends or unusual records.

Prepare features consistently before model fitting

The held-out test set evaluates records that were not used to fit the model.

01 Inspect & clean
Review the CSV, data types, missing values and duplicates.

>

02 Separate inputs
Use four numeric features and one categorical location field.

>

03 Encode category
OneHotEncoder turns location groups into indicator columns.

>

04 Split records
80% train and 20% test; random_state=42 for reproducibility.

>

05 Fit pipeline
Transform inputs and fit Linear Regression on training data.

SCIKIT-LEARN PIPELINE

```
ColumnTransformer + OneHotEncoder(handle_unknown="ignore") + LinearRegression
```

Numeric features are passed through; the pipeline reuses the same transformation during training and prediction.

From raw records to price predictions

Linear Regression is a transparent baseline for a continuous target such as listed house price.

DATA
CSV records

>

INSPECT
Pandas / EDA

>

TRANSFORM
Numeric + encoded location

>

TRAIN
Linear Regression

>

PREDICT
Estimated INR

LINEAR MODEL (CONCEPT)

Predicted price = intercept + sum(feature coefficient * feature value)

WHY THIS BASELINE?

Simple to explain; coefficients show fitted relationships, not causal effects.

After fitting: predict test rows -> compare with known prices -> calculate metrics -> inspect diagnostic plots.

Reported test-set metrics

Values below are reported for the synthetic dataset and must not be interpreted as real-market accuracy.

MAE

INR 382,443

Average absolute prediction error; same units as price.

RMSE

INR 490,665

Square-root mean squared error; larger errors weigh more.

R-squared

0.985

Proportion of test-target variation explained by the fitted model.

INTERPRETATION

MAE/RMSE express error in INR. R-squared summarizes fit on this held-out sample; none alone proves generalization.

Test subset: 100 records | Split: 80:20 | random_state=42

Use plots to inspect patterns and errors

The report describes these diagnostic plots; no plot values or individual observations are inferred here.

01 Area vs Listed Price

Scatter plot: inspect the general size-price trend and spread; area alone does not explain every difference.

02 Actual vs Predicted

Compare test values with predictions; a 45-degree reference line indicates perfect agreement.

03 Error Distribution

Histogram of actual minus predicted prices; values near zero indicate smaller errors.

04 Linear Coefficients

Compare fitted feature weights; signs show model direction, but not causation.

READ WITH CARE **Visual patterns describe this synthetic sample; coefficients depend on units and encoding.**

An educational baseline - not a valuation service

A practical tool would need more representative data, richer features and careful validation.

LIMITATIONS

- Synthetic data, not verified market transactions
- Broad location categories; no exact neighbourhood
- Missing condition, amenities, floor, legal status and demand
- Linear Regression may miss nonlinear patterns
- Results can shift with data and market changes

FUTURE SCOPE

- Use larger, verified listing or transaction data
- Add locality, amenities and property condition
- Compare tree-based models with Linear Regression
- Use cross-validation and systematic evaluation
- Report prediction ranges; consider Streamlit / Flask

RESPONSIBLE USE **Treat any output as an estimate; validate locally and communicate uncertainty.**

CONCLUSION

A clear starting point for regression-based prediction

The project demonstrates the supervised-learning workflow: prepare and explore data, encode location, split records, train Linear Regression, predict held-out prices, and evaluate results.

TAKEAWAY

The synthetic results teach the method - not real-world market accuracy.

Real use requires representative data, validation and uncertainty.

REFERENCES

01 Scikit-learn User Guide - Linear Models: scikit-learn.org/stable/modules/linear_model.html

02 Scikit-learn - OneHotEncoder: scikit-learn.org/stable/modules/generated/sklearn.preprocessing.OneHotEncoder.html

03 Pandas documentation: pandas.pydata.org/docs/

04 Matplotlib documentation: matplotlib.org/stable/

BCA (Artificial Intelligence), Section A | Mentor: Harsh Shrivastav